

Year 7 · Chapter 2 · Angles: answers (teacher copy)

2.01 Naming angles (booklet p. 3)

WE DO examples (teacher)

1. Ex 1a: $\angle ABC$ or $\angle CBA$
2. Ex 1b: vertex K in the middle
3. Ex 2a: $\angle QOR$
4. Ex 2b: $\angle QOR$, $\angle ROP$, $\angle QOP$
5. Ex 3a: $\angle LON$
6. Ex 3b: There are several angles at B, so we need 3 letters

Easy · Set A

1. $\angle PQR$
2. $\angle DEF$
3. $\angle XYZ$
4. $\angle MNK$
5. $\angle STU$
6. $\angle GHJ$

Easy · Set A2 (quick drill)

1. B
2. Q
3. Y
4. E
5. N
6. T
7. H
8. K

Easy · Set B

1. vertex P, arms PF and PR
2. vertex Z, arms ZM and ZQ

Medium · Set C

1. $\angle DBE$
2. $\angle DBE$
3. $\angle CBA$
4. 6
5. $\angle DBA$
6. $\angle CBE$

Challenging · Set D

1. $\angle LMN$
2. $\angle QRS$
3. Yes: the vertex B is in the middle of both
4. e.g. two angles sharing arm QR

Skill drill 1 (Easy basics)

- | | | |
|-------|-------|----------------------------------|
| 1. X | 11. S | 21. $\angle VSB$ or $\angle BSV$ |
| 2. D | 12. A | 22. $\angle WEF$ or $\angle FEW$ |
| 3. D | 13. Y | 23. $\angle QEF$ or $\angle FEQ$ |
| 4. B | 14. F | 24. $\angle SXT$ or $\angle TXS$ |
| 5. Y | 15. G | 25. $\angle PXV$ or $\angle VXP$ |
| 6. N | 16. W | 26. $\angle CRP$ or $\angle PRC$ |
| 7. E | 17. L | 27. $\angle EGZ$ or $\angle ZGE$ |
| 8. W | 18. P | 28. $\angle LPZ$ or $\angle ZPL$ |
| 9. D | 19. V | 29. $\angle DPE$ or $\angle EPD$ |
| 10. V | 20. V | |

Skill drill 2 (Medium basics)

- | | | |
|--------------|---------------|---------------|
| 1. ZM and ZP | 7. NE and NU | 13. WL and WE |
| 2. CM and CS | 8. AS and AK | 14. VZ and VM |
| 3. NU and NX | 9. QC and QN | 15. RQ and RY |
| 4. QY and QH | 10. PV and PD | 16. WX and WS |
| 5. PV and PZ | 11. UL and UY | 17. TS and TK |
| 6. LE and LM | 12. YE and YG | 18. SZ and ST |

- | | | |
|---------------|-----------|-----------|
| 19. WU and WL | 24. false | 29. true |
| 20. FY and FQ | 25. true | 30. true |
| 21. false | 26. true | 31. false |
| 22. true | 27. true | 32. false |
| 23. true | 28. true | |

Extension

1. 10
2. $\angle AOB$, $\angle AOC$, $\angle AOD$, $\angle BOC$, $\angle BOD$, $\angle COD$

Exit ticket

1. Level 1: $\angle TUV$
2. Level 2: U, because it is the point where the arms meet
3. Level 3: two angles sharing arm BC

2.02 Measuring and drawing angles

(booklet p. 11)

WE DO examples (teacher)

1. Ex 1a: 64°
2. Ex 1b: accurate 84° angle
3. Ex 2a: 128°
4. Ex 2b: accurate 117° angle
5. Ex 3a: draw 150° on the other side
6. Ex 3b: draw 60° on the other side

Easy · Set A

1. 35°
2. 72°
3. 90°
4. 105°
5. 150°
6. 48°

Easy · Set A2 (quick drill)

1. 150°
2. 60°
3. 165°
4. 90°
5. 110°
6. 30°
7. 175°
8. 120°

Easy · Set B

1. 41° drawn
2. 150° drawn

Medium · Set C

1. 96° drawn
2. 23° drawn
3. 165° drawn
4. 72° drawn

Challenging · Set D

1. $360 - 195 = 165^\circ$ on the other side
2. three-quarter turn
3. 230°
4. 215°

Skill drill 1 (Easy basics)

- | | | |
|----------------|-----------------|-----------------|
| 1. 15° | 6. 140° | 11. 65° |
| 2. 155° | 7. 140° | 12. 40° |
| 3. 160° | 8. 135° | 13. 155° |
| 4. 170° | 9. 85° | 14. 105° |
| 5. 130° | 10. 140° | 15. 115° |

- | | | |
|-----------------|-----------------|-----------------|
| 16. 95° | 21. 20° | 26. 165° |
| 17. 70° | 22. 120° | 27. 5° |
| 18. 35° | 23. 15° | 28. 175° |
| 19. 145° | 24. 75° | |
| 20. 170° | 25. 160° | |

Skill drill 2 (Medium basics)

- | | | |
|----------------|-----------------|------------|
| 1. 48° | 11. 174° | 21. reflex |
| 2. 103° | 12. 62° | 22. acute |
| 3. 101° | 13. reflex | 23. obtuse |
| 4. 30° | 14. obtuse | 24. reflex |
| 5. 81° | 15. obtuse | 25. obtuse |
| 6. 78° | 16. reflex | 26. reflex |
| 7. 49° | 17. right | 27. acute |
| 8. 138° | 18. obtuse | 28. reflex |
| 9. 99° | 19. reflex | |
| 10. 55° | 20. reflex | |

Extension

- 150° and 90°
- 75°

Exit ticket

- Level 1: 58°
- Level 2: accurate 135°
- Level 3: draw 110° the other way

2.03 Classifying angles (booklet p. 19)

WE DO examples (teacher)

- Ex 1a: acute, right, obtuse, straight, reflex, revolution
- Ex 1b: $\angle WIH$, obtuse
- Ex 2a: obtuse
- Ex 2b: 270° , reflex
- Ex 3a: $\angle DRA$ acute (70°), $\angle ARB$ obtuse (110°)
- Ex 3b: No: 90° + more than 90° is already more than 180°

Easy · Set A (quick drill)

- | | | |
|-------------|----------------|------------|
| 1. acute | 7. acute | 13. obtuse |
| 2. right | 8. obtuse | 14. reflex |
| 3. obtuse | 9. reflex | 15. reflex |
| 4. straight | 10. revolution | 16. acute |
| 5. reflex | 11. acute | |
| 6. acute | 12. obtuse | |

Easy · Set B

- acute
- obtuse
- reflex
- revolution
- obtuse
- reflex

Medium · Set C

- $\angle GLU$ acute
- $\angle VRP$ obtuse
- $\angle PNE$ right
- $\angle SMA$ reflex
- $\angle ZXY$ straight
- $\angle MVT$ acute

Challenging · Set D

- 120° , obtuse
- Yes, e.g. $60^\circ + 50^\circ = 110^\circ$
- 100° , 125° , 150° , 175°
- 19 (201° to 219°)

Skill drill 1 (Easy basics)

- | | | |
|-----------|-----------|-----------|
| 1. reflex | 2. obtuse | 3. reflex |
|-----------|-----------|-----------|

- | | | |
|--------------|------------|------------|
| 4. reflex | 13. reflex | 22. obtuse |
| 5. reflex | 14. obtuse | 23. obtuse |
| 6. right | 15. obtuse | 24. obtuse |
| 7. reflex | 16. obtuse | 25. obtuse |
| 8. acute | 17. reflex | 26. obtuse |
| 9. reflex | 18. reflex | 27. acute |
| 10. reflex | 19. acute | 28. right |
| 11. straight | 20. reflex | 29. acute |
| 12. obtuse | 21. acute | |

Skill drill 2 (Medium basics)

- | | | |
|--------------------------|--------------------------|-----------|
| 1. 225° , reflex | 11. 160° , obtuse | 21. true |
| 2. 305° , reflex | 12. 145° , obtuse | 22. true |
| 3. 100° , obtuse | 13. 205° , reflex | 23. true |
| 4. 175° , obtuse | 14. 225° , reflex | 24. false |
| 5. 260° , reflex | 15. 285° , reflex | 25. false |
| 6. 215° , reflex | 16. 340° , reflex | 26. true |
| 7. 105° , obtuse | 17. true | 27. true |
| 8. 295° , reflex | 18. false | 28. true |
| 9. 160° , obtuse | 19. true | |
| 10. 220° , reflex | 20. true | |

Extension

- 900°
- 3 revolutions

Exit ticket

- Level 1: acute
- Level 2: reflex; straight
- Level 3: $\angle DRA$ acute; $\angle ARB$ obtuse

2.04 Complementary and supplementary angles (booklet p. 27)

WE DO examples (teacher)

- Ex 1a: 55° ; 12°
- Ex 1b: 135° ; 80°
- Ex 2a: $m = 62$
- Ex 2b: $k = 133$
- Ex 3a: $x = 29$
- Ex 3b: 90° each

Easy · Set A (quick drill)

- | | | |
|---------------|-----------------|-----------------|
| 1. 55° | 7. 65° | 13. 8° |
| 2. 12° | 8. 22° | 14. 116° |
| 3. 86° | 9. 78° | 15. 90° |
| 4. 30° | 10. 9° | 16. 45° |
| 5. 45° | 11. 150° | |
| 6. 1° | 12. 65° | |

Easy · Set B

- 135°
- 80°
- 2°
- 90°
- 168°
- 45°

Medium · Set C

- $m = 62$
- $k = 133$
- $a = 29$
- $y = 58$
- $p = 15$
- $b = 90$

Challenging · Set D

- 55° and 35°
- 36°

3. $x = 40$
4. No: it is already more than 90°

Skill drill 1 (Easy basics)

- | | | |
|----------------|-----------------|-----------------|
| 1. 76° | 15. 6° | 29. 126° |
| 2. 56° | 16. 32° | 30. 14° |
| 3. 53° | 17. 72° | 31. 21° |
| 4. 54° | 18. 79° | 32. 19° |
| 5. 24° | 19. 16° | 33. 15° |
| 6. 41° | 20. 5° | 34. 127° |
| 7. 84° | 21. 73° | 35. 141° |
| 8. 36° | 22. 132° | 36. 167° |
| 9. 67° | 23. 103° | 37. 99° |
| 10. 70° | 24. 162° | 38. 118° |
| 11. 10° | 25. 179° | 39. 67° |
| 12. 89° | 26. 55° | 40. 90° |
| 13. 68° | 27. 106° | |
| 14. 47° | 28. 2° | |

Skill drill 2 (Medium basics)

- | | | |
|-------|--------------|---------------|
| 1. C | 13. N | 25. $x = 50$ |
| 2. N | 14. S | 26. $m = 64$ |
| 3. C | 15. S | 27. $p = 46$ |
| 4. N | 16. N | 28. $k = 75$ |
| 5. C | 17. C | 29. $x = 87$ |
| 6. C | 18. S | 30. $x = 117$ |
| 7. S | 19. N | 31. $y = 53$ |
| 8. C | 20. N | 32. $k = 107$ |
| 9. S | 21. $x = 94$ | 33. $x = 62$ |
| 10. S | 22. $x = 28$ | 34. $k = 16$ |
| 11. C | 23. $k = 23$ | 35. $p = 106$ |
| 12. S | 24. $m = 13$ | |

Extension

1. 60°
2. 40°

Exit ticket

1. Level 1: 28°
2. Level 2: 62°
3. Level 3: $x = 45$

2.05 Angles at a point and vertically opposite angles (booklet p. 35)

WE DO examples (teacher)

1. Ex 1a: $a = 110$
2. Ex 1b: $m = 25$
3. Ex 2a: $a = 130$
4. Ex 2b: $x = 120$
5. Ex 3a: $p = 145$
6. Ex 3b: $p = 155, q = 25, r = 155$

Easy · Set A

1. $y = 145$
2. $a = 80$
3. $m = 44$
4. $x = 98$
5. $k = 25$
6. $d = 90$

Easy · Set A2 (quick drill)

1. 125°
2. 144°
3. 75°
4. 120°
5. 67°
6. 118°

7. 72°
8. 116°

Easy · Set B

1. $y = 35$
2. $m = 80$
3. $f = 60$

Medium · Set C

1. $f = 110$
2. $t = 83$
3. $x = 90$

Challenging · Set D

1. $x = 132$ (straight line), $y = 48$ (vertically opposite)
2. $t = 173$ (angles at a point)
3. 120° each
4. $115^\circ, 65^\circ, 115^\circ$

Skill drill 1 (Easy basics)

- | | | |
|---------------|---------------|---------------|
| 1. $a = 70$ | 13. $a = 52$ | 25. $a = 217$ |
| 2. $a = 99$ | 14. $a = 60$ | 26. $a = 28$ |
| 3. $a = 150$ | 15. $a = 88$ | 27. $a = 174$ |
| 4. $a = 102$ | 16. $a = 101$ | 28. $a = 26$ |
| 5. $a = 159$ | 17. $a = 119$ | 29. $a = 188$ |
| 6. $a = 164$ | 18. $a = 77$ | 30. $a = 202$ |
| 7. $a = 162$ | 19. $a = 156$ | 31. $a = 62$ |
| 8. $a = 154$ | 20. $a = 78$ | 32. $a = 221$ |
| 9. $a = 39$ | 21. $a = 103$ | 33. $a = 123$ |
| 10. $a = 116$ | 22. $a = 132$ | 34. $a = 155$ |
| 11. $a = 84$ | 23. $a = 202$ | 35. $a = 179$ |
| 12. $a = 128$ | 24. $a = 85$ | 36. $a = 169$ |

Skill drill 2 (Medium basics)

- | | | |
|----------------------------------|-----------------------------------|---------------|
| 1. $v = 77^\circ, n = 103^\circ$ | 9. $v = 70^\circ, n = 110^\circ$ | 19. $a = 143$ |
| 2. $v = 84^\circ, n = 96^\circ$ | | 20. $a = 130$ |
| 3. $v = 76^\circ, n = 104^\circ$ | 10. $v = 61^\circ, n = 119^\circ$ | 21. $a = 104$ |
| 4. $v = 127^\circ, n = 53^\circ$ | 11. $v = 91^\circ, n = 89^\circ$ | 22. $a = 120$ |
| 5. $v = 130^\circ, n = 50^\circ$ | 12. $v = 136^\circ, n = 44^\circ$ | 23. $a = 126$ |
| 6. $v = 83^\circ, n = 97^\circ$ | 13. $a = 153$ | 24. $a = 70$ |
| 7. $v = 34^\circ, n = 146^\circ$ | 14. $a = 30$ | 25. $a = 71$ |
| 8. $v = 74^\circ, n = 106^\circ$ | 15. $a = 33$ | 26. $a = 74$ |
| | 16. $a = 63$ | 27. $a = 91$ |
| | 17. $a = 141$ | 28. $a = 73$ |
| | 18. $a = 140$ | 29. $a = 82$ |

Extension

1. $30^\circ, 60^\circ, 90^\circ$
2. $x = 30$

Exit ticket

1. Level 1: 116
2. Level 2: 38°
3. Level 3: 135

2.06 Parallel and perpendicular lines

(booklet p. 43)

WE DO examples (teacher)

1. Ex 1a: $AB \parallel DC$ (or $AD \parallel BC$); $AB \perp BC$
2. Ex 1b: AB is perpendicular to CD ; EF is parallel to GH
3. Ex 2a: construction
4. Ex 2b: construction
5. Ex 3a: construction
6. Ex 3b: e.g. shelves ($\parallel$), wall and floor ($\perp$)

Easy · Set A (quick drill)

1. DC

2. BC or AD
3. AD
4. No (not in a rectangle unless it is a square)
5. at B
6. AD $\parallel$ BC

Easy · Set B

1. True
2. True
3. False
4. True

Medium · Set C

1. construction
2. construction

Challenging · Set D

1. construction
2. square construction

Skill drill 1 (Easy basics)

- | | | |
|----------------------|----------------------|-------------|
| 1. PQ $\perp$ RS | 9. AB $\parallel$ CD | 17. $\perp$ |
| 2. $\perp$ | 10. EF $\perp$ GH | 18. $\perp$ |
| 3. TU $\perp$ VW | 11. $\parallel$ | 19. P |
| 4. $\perp$ | 12. $\parallel$ | 20. P |
| 5. XY $\parallel$ ZW | 13. $\perp$ | 21. P |
| 6. KL $\parallel$ MN | 14. $\perp$ | 22. N |
| 7. $\parallel$ | 15. P | 23. N |
| 8. $\perp$ | 16. P | 24. N |

Skill drill 2 (Medium basics)

- | | | |
|----------|---------------|---------------|
| 1. true | 9. false | 17. ruler |
| 2. true | 10. true | 18. compasses |
| 3. true | 11. true | 19. compasses |
| 4. false | 12. false | 20. ruler |
| 5. true | 13. both | 21. both |
| 6. true | 14. compasses | 22. both |
| 7. true | 15. ruler | |
| 8. false | 16. both | |

Extension

1. construction
2. every point on it is the same distance from A and B

Exit ticket

1. Level 1: $\parallel$
2. Level 2: e.g. wall and floor
3. Level 3: construction

2.07 Corresponding angles (booklet p. 51)

WE DO examples (teacher)

1. Ex 1a: bottom line, upper right
2. Ex 1b: $\alpha = 65$
3. Ex 2a: $m = 110$
4. Ex 2b: the angles sit in an F
5. Ex 3a: $x = 125$, $y = 55$
6. Ex 3b: Only when the lines are parallel

Easy · Set A

1. bottom ul
2. top lr
3. bottom ll
4. top ur
5. bottom lr
6. top ul

Easy · Set A2 (quick drill)

1. 65°
2. 110°

3. 38°
4. 142°
5. 90°
6. 75°
7. 120°
8. 51°

Easy · Set B

1. $\alpha = 65$
2. $k = 50$
3. $d = 110$

Medium · Set C

1. $b = 35$
2. $m = 115$
3. $q = 118$ (corresponding 62° , then straight line)
4. $x = 122$
5. $y = 75$ (vertically opposite, then corresponding)
6. $t = 140$

Challenging · Set D

1. a line crossing two or more lines
2. drawing
3. No
4. angles on a straight line

Skill drill 1 (Easy basics)

- | | | |
|-----------------|--------------------|--------------------|
| 1. 89° | 11. 64° | 21. $\alpha = 140$ |
| 2. 104° | 12. 121° | 22. $\alpha = 80$ |
| 3. 23° | 13. 133° | 23. $\alpha = 65$ |
| 4. 136° | 14. 124° | 24. $\alpha = 150$ |
| 5. 93° | 15. 128° | 25. $\alpha = 65$ |
| 6. 55° | 16. 94° | 26. $\alpha = 145$ |
| 7. 100° | 17. $\alpha = 90$ | 27. $\alpha = 45$ |
| 8. 44° | 18. $\alpha = 145$ | 28. $\alpha = 30$ |
| 9. 28° | 19. $\alpha = 65$ | |
| 10. 143° | 20. $\alpha = 100$ | |

Skill drill 2 (Medium basics)

- | | | |
|-------------------------------|--------------|--------------|
| 1. $\alpha = 70$, $b = 110$ | 13. $x = 32$ | 25. $x = 15$ |
| 2. $\alpha = 85$, $b = 95$ | 14. $x = 14$ | 26. $x = 40$ |
| 3. $\alpha = 90$, $b = 90$ | 15. $x = 32$ | 27. $x = 34$ |
| 4. $\alpha = 150$, $b = 30$ | 16. $x = 16$ | 28. $x = 19$ |
| 5. $\alpha = 35$, $b = 145$ | 17. $x = 9$ | 29. no |
| 6. $\alpha = 135$, $b = 45$ | 18. $x = 27$ | 30. no |
| 7. $\alpha = 145$, $b = 35$ | 19. $x = 11$ | 31. yes |
| 8. $\alpha = 60$, $b = 120$ | 20. $x = 16$ | 32. yes |
| 9. $\alpha = 105$, $b = 75$ | 21. $x = 39$ | 33. no |
| 10. $\alpha = 30$, $b = 150$ | 22. $x = 27$ | 34. no |
| 11. $\alpha = 125$, $b = 55$ | 23. $x = 6$ | 35. yes |
| 12. $\alpha = 40$, $b = 140$ | 24. $x = 7$ | 36. no |

Extension

1. $y = 40$
2. $\alpha = 30$

Exit ticket

1. Level 1: 60
2. Level 2: an F
3. Level 3: $x = 20$

2.08 Alternate angles (booklet p. 59)

WE DO examples (teacher)

1. Ex 1a: bottom line, upper right
2. Ex 1b: $x = 60$
3. Ex 2a: $k = 115$
4. Ex 2b: True
5. Ex 3a: $\alpha = 48$, $b = 132$

6. Ex 3b: the angles sit inside a Z

Easy · Set A

1. bottom ul
2. top ll
3. bottom ur
4. top lr
5. bottom ul
6. top ll

Easy · Set A2 (quick drill)

1. 60°
2. 115°
3. 48°
4. 132°
5. 130°
6. 50°
7. 72°
8. 108°

Easy · Set B

1. $x = 70$
2. $p = 125$
3. $w = 38$

Medium · Set C

1. $c = 65$
2. $t = 112$
3. $d = 130$ (alternate 50° , straight line)
4. $e = 73$ (corresponding)
5. $g = 140$
6. $h = 100$ (alternate 80° , then straight line)

Challenging · Set D

1. drawing
2. $x = 45$
3. alternate angles and angles on a straight line
4. It is perpendicular to both lines ($y = 45$)

Skill drill 1 (Easy basics)

- | | | |
|----------------|-----------------|---------------|
| 1. 111° | 11. 133° | 21. $a = 140$ |
| 2. 28° | 12. 68° | 22. $a = 45$ |
| 3. 78° | 13. 122° | 23. $a = 40$ |
| 4. 119° | 14. 138° | 24. $a = 150$ |
| 5. 91° | 15. 116° | 25. $a = 115$ |
| 6. 75° | 16. 142° | 26. $a = 155$ |
| 7. 105° | 17. $a = 90$ | 27. $a = 75$ |
| 8. 139° | 18. $a = 135$ | 28. $a = 40$ |
| 9. 113° | 19. $a = 90$ | |
| 10. 42° | 20. $a = 85$ | |

Skill drill 2 (Medium basics)

- | | | |
|--------------------------------------|--------------------------------------|--------------|
| 1. $a = 83^\circ$, $c = 97^\circ$ | 11. $a = 30^\circ$, $c =$ | 21. $x = 27$ |
| 2. $a = 13^\circ$, $c = 167^\circ$ | 150° | 22. $x = 12$ |
| 3. $a = 117^\circ$, $c = 63^\circ$ | 12. $a = 23^\circ$, $c = 157^\circ$ | 23. $x = 38$ |
| 4. $a = 43^\circ$, $c = 137^\circ$ | 13. $a = 51^\circ$, $c = 129^\circ$ | 24. $x = 19$ |
| 5. $a = 109^\circ$, $c = 71^\circ$ | 14. $a = 50^\circ$, $c =$ | 25. $x = 30$ |
| 6. $a = 52^\circ$, $c = 128^\circ$ | 130° | 26. $x = 24$ |
| 7. $a = 62^\circ$, $c = 118^\circ$ | 15. $a = 39^\circ$, $c = 141^\circ$ | 27. $x = 27$ |
| 8. $a = 151^\circ$, $c = 29^\circ$ | 16. $x = 33$ | 28. $x = 39$ |
| 9. $a = 70^\circ$, $c =$ | 17. $x = 14$ | 29. $x = 9$ |
| 110° | 18. $x = 28$ | 30. $x = 18$ |
| 10. $a = 37^\circ$, $c = 143^\circ$ | 19. $x = 9$ | 31. $x = 37$ |
| | 20. $x = 27$ | |

Extension

1. $a = 25$
2. $m = 35$; both 105°

Exit ticket

1. Level 1: 65
2. Level 2: Z
3. Level 3: $x = 35$

2.09 Co-interior angles (booklet p. 67)

WE DO examples (teacher)

1. Ex 1a: bottom line, upper right
2. Ex 1b: $y = 60$
3. Ex 2a: $m = 105$
4. Ex 2b: No: they add to 180° (unless both are 90°)
5. Ex 3a: $p = 65$, $q = 115$
6. Ex 3b: corresponding (equal), alternate (equal), co-interior (add to 180°)

Easy · Set A

1. bottom ul
2. top lr
3. bottom ur
4. top ll
5. top lr
6. bottom ul

Easy · Set A2 (quick drill)

1. 60°
2. 105°
3. 85°
4. 120°
5. 35°
6. 150°
7. 70°
8. 92°

Easy · Set B

1. $a = 60$
2. $x = 110$
3. $b = 145$

Medium · Set C

1. $q = 62$
2. $k = 80$
3. $z = 123$ (co-interior 57° , then straight line)
4. $m = 64$ (corresponding)
5. $x = 130$
6. $y = 110$ (corresponding)

Challenging · Set D

1. $x = 60$
2. 90° each; it is perpendicular to the parallel lines
3. drawing
4. $x = 60$

Skill drill 1 (Easy basics)

- | | | |
|-----------------|-----------------|---------------|
| 1. 31° | 11. 100° | 21. $a = 60$ |
| 2. 47° | 12. 137° | 22. $a = 100$ |
| 3. 79° | 13. 135° | 23. $a = 95$ |
| 4. 108° | 14. 26° | 24. $a = 90$ |
| 5. 74° | 15. 38° | 25. $a = 85$ |
| 6. 104° | 16. 62° | 26. $a = 65$ |
| 7. 112° | 17. $a = 145$ | 27. $a = 70$ |
| 8. 101° | 18. $a = 110$ | 28. $a = 95$ |
| 9. 91° | 19. $a = 30$ | |
| 10. 157° | 20. $a = 125$ | |

Skill drill 2 (Medium basics)

- | | | |
|-------------|-------------|-------------|
| 1. $y = 48$ | 4. $y = 21$ | 7. $y = 15$ |
| 2. $y = 16$ | 5. $y = 50$ | 8. $y = 55$ |
| 3. $y = 32$ | 6. $y = 58$ | 9. $y = 49$ |

- | | | |
|--------------|---------|---------|
| 10. $y = 54$ | 18. no | 26. no |
| 11. $y = 40$ | 19. yes | 27. no |
| 12. $y = 43$ | 20. no | 28. no |
| 13. $y = 8$ | 21. no | 29. no |
| 14. $y = 51$ | 22. no | 30. no |
| 15. $y = 5$ | 23. no | 31. yes |
| 16. $y = 23$ | 24. yes | 32. yes |
| 17. yes | 25. yes | |

Extension

1. 80° and 100°
2. $a = 50$

Exit ticket

1. Level 1: 70
2. Level 2: 180°
3. Level 3: $y = 20$

2.10 Finding unknown angles on parallel lines (booklet p. 75)

WE DO examples (teacher)

1. Ex 1a: $a = 65$ (corresponding then straight line; or co-interior with 115° pair)
2. Ex 1b: $m = 35$ (alternate)
3. Ex 2a: $x = 37$ (corr.), $y = 143$ (straight line)
4. Ex 2b: $k = 65$ (alternate)
5. Ex 3a: $p = 62$, $m = 62$, $z = 118$
6. Ex 3b: use two rules in steps, e.g. vertically opposite then alternate

Easy · Set A

1. $a = 65$
2. $b = 35$
3. $k = 65$
4. $d = 125$
5. $x = 50$
6. $q = 118$

Easy · Set A2 (quick drill)

1. corresponding, equal
2. alternate, equal
3. co-interior, add to 180°
4. corresponding, equal
5. alternate, equal
6. co-interior, add to 180°

Easy · Set B

1. $y = 132$ (alternate, then straight line)
2. $t = 110$
3. $x = 123$

Medium · Set C

1. $p = 62$, $m = 62$, $z = 118$
2. $t = 68$, $d = 112$, $a = 68$
3. $x = 130$, $y = 50$

Challenging · Set D

1. four of 57° and four of 123°
2. 135° and 45°
3. The lines are parallel so each angle has a matching corresponding angle
4. any correct question

Skill drill 1 (Easy basics)

- | | | |
|------------------|----------------|-------------------|
| 1. co-interior | 5. co-interior | 9. co-interior |
| 2. co-interior | 6. alternate | 10. corresponding |
| 3. alternate | 7. alternate | 11. corresponding |
| 4. corresponding | 8. alternate | 12. corresponding |

- | | | |
|---------------|--------------|---------------|
| 13. $a = 100$ | 16. $a = 80$ | 19. $a = 85$ |
| 14. $a = 105$ | 17. $a = 30$ | 20. $a = 150$ |
| 15. $a = 150$ | 18. $a = 85$ | 21. $a = 135$ |

Skill drill 2 (Medium basics)

- | | | |
|---------------------------------|---------------------------------|--------------------------------------|
| 1. $a = 155$
(corresponding) | 7. $a = 100$ (co-interior) | 13. $129^\circ, 51^\circ, 51^\circ$ |
| 2. $a = 40$ (co-interior) | 8. $a = 55$ (co-interior) | 14. $52^\circ, 128^\circ, 128^\circ$ |
| 3. $a = 135$ (co-interior) | 9. $a = 110$ (co-interior) | 15. $55^\circ, 125^\circ, 125^\circ$ |
| 4. $a = 145$
(alternate) | 10. $a = 50$ (co-interior) | 16. $31^\circ, 149^\circ, 149^\circ$ |
| 5. $a = 150$ (co-interior) | 11. $a = 85$
(corresponding) | 17. $79^\circ, 101^\circ, 101^\circ$ |
| 6. $a = 65$
(corresponding) | 12. $a = 95$
(alternate) | 18. $104^\circ, 76^\circ, 76^\circ$ |
| | | 19. $142^\circ, 38^\circ, 38^\circ$ |
| | | 20. $18^\circ, 162^\circ, 162^\circ$ |
| | | 21. $130^\circ, 50^\circ, 50^\circ$ |
| | | 22. $16^\circ, 164^\circ, 164^\circ$ |
| | | 23. $127^\circ, 53^\circ, 53^\circ$ |
| | | 24. $141^\circ, 39^\circ, 39^\circ$ |

Extension

1. $x = 16$
2. $x = 40$

Exit ticket

1. Level 1: 58
2. Level 2: alternate (vertically opposite then corresponding)
3. Level 3: four 58° and four 122°

2.11 Proving lines are parallel (booklet p. 83)

WE DO examples (teacher)

1. Ex 1a: Yes: co-interior angles $135 + 45 = 180$
2. Ex 1b: No: $110 + 72 = 182$
3. Ex 2a: Yes: corresponding angles equal
4. Ex 2b: $x = 95$
5. Ex 3a: corresponding equal, alternate equal, co-interior supplementary
6. Ex 3b: we use the angle to prove the lines are parallel (the reverse of the rule)

Easy · Set A (quick drill)

- | | | |
|--------|--------|---------|
| 1. yes | 5. yes | 9. no |
| 2. no | 6. no | 10. yes |
| 3. yes | 7. yes | 11. no |
| 4. no | 8. yes | 12. yes |

Easy · Set B

1. Yes (corresponding equal)
2. No (alternate $50 \neq 52$)
3. Yes (co-interior $115 + 65 = 180$)

Medium · Set C

1. 112
2. 47
3. 85
4. 55
5. 70
6. 37

Challenging · Set D

1. alternate angles equal, so parallel
2. $108 + 72 = 180$, co-interior supplementary, so parallel
3. No: corresponding angles are not equal
4. e.g. corresponding angles 60° and 70°

Skill drill 1 (Easy basics)

- | | | |
|--------|--------|---------|
| 1. yes | 5. no | 9. no |
| 2. yes | 6. yes | 10. no |
| 3. yes | 7. yes | 11. yes |
| 4. yes | 8. no | 12. no |

- | | | | | | |
|---------|---------|---------|-----------------|-----------------|-------------------|
| 13. yes | 17. yes | 21. no | 22. co-interior | 24. alternate | 26. alternate |
| 14. no | 18. no | 22. no | angles are | angles are | angles are |
| 15. no | 19. yes | 23. yes | supplementary | equal | equal |
| 16. yes | 20. yes | 24. yes | 23. alternate | 25. co-interior | 27. corresponding |
| | | | angles are | angles are | angles are |
| | | | equal | supplementary | equal |

Skill drill 2 (Medium basics)

- | | | |
|--------------|-----------------|-----------------|
| 1. $x = 150$ | 12. $x = 150$ | 18. co-interior |
| 2. $x = 45$ | 13. $x = 40$ | angles are |
| 3. $x = 80$ | 14. $x = 125$ | supplementary |
| 4. $x = 125$ | 15. $x = 60$ | 19. co-interior |
| 5. $x = 85$ | 16. alternate | angles are |
| 6. $x = 125$ | angles are | supplementary |
| 7. $x = 35$ | equal | 20. co-interior |
| 8. $x = 30$ | 17. co-interior | angles are |
| 9. $x = 120$ | angles are | supplementary |
| 10. $x = 65$ | supplementary | 21. alternate |
| 11. $x = 35$ | | angles are |
| | | equal |

Extension

- $x = 132$; $132 + 48 = 180$, so yes (co-interior)
- Measurements are never exact: a proof uses the angle rules

Exit ticket

- Level 1: Yes
- Level 2: No: $105 + 85 = 190$, not 180
- Level 3: $x = 37$